Abraham Khan

Email: awkhan3@ncsu.edu • Citizenship: United States • Status: PhD Candidate in Mathematics

Overview

As a computational mathematician, I develop fast algorithms for densely structured linear and multilinear algebra problems. My current research focuses on developing tensor-based algorithms with an eye towards applications in scientific computing and machine learning; in particular, my focus is on efficiently computing kernel matrices. My broader vision is to develop efficient numerical methods that scale efficiently for higher problem sizes and mitigate the curse of dimensionality: the ultimate goal is for the complexity to grow linearly with the problem's intrinsic dimension and overall size.

Research Interests

Numerical linear algebra; multilinear algebra; machine learning; approximation theory; theoretical computer science.

Education

Ph.D., Mathematics, North Carolina State University *2020–Expected March 2026*
Raleigh, NC

Dissertation: Parametric kernel approximations using tensors.

Advisor: Dr. Arvind K. Saibaba.

Qualifying exams: Matrix Theory; Functional Analysis; Abstract Algebra.

B.S., Mathematics, University of South Carolina *2014–2018*
Columbia, SC

B.S., Computer Science, University of South Carolina *2014–2018*
Columbia, SC

Research Experience & Appointments

Research Assistant, North Carolina State University *2022–Present*
Raleigh, NC

Advisor: Dr. Arvind K. Saibaba

NSF RTG Trainee in Randomized Numerical Analysis

Work supported by NSF grants (DMS-1745654, DMS-1845406, DMS-2411198)

- Developed the *Parametric Tensor Train Kernel (PTTK)*, a tensor-train–based approach for computing parametric low-rank approximations of parametric matrices across the full parameter domain.
- Developed *parametric hierarchical matrices*, a new class of hierarchical matrices that efficiently approximates parametric kernel matrices across the full parameter domain.
- Currently investigating hierarchical matrix approximation using tensor decompositions.

Givens Intern, Argonne National Laboratory *Summer 2024*
Lemont, IL

Mentor: Dr. Vishwas Rao

- Explored data-sparse approximations (hierarchical matrices) and developed the concept of parametric hierarchical matrices.
- Continued this work after the internship to further develop the parametric hierarchical matrix framework.

Undergraduate Researcher, University of South Carolina *2016–2019*
Columbia, SC

Mentor: Dr. Joshua Cooper

- Explored a problem that is at the intersection of order theory and theoretical computer science: determine the computational complexity of counting the number of linear extensions of a dimension two semi-order.

Publications

- **Khan, A.**; Saibaba, A. K. (2025). *Parametric kernel low-rank approximations using tensor train decomposition. SIAM Journal on Matrix Analysis and Applications*, **46**(2), 1006–1036. (33 pages).
- **Khan, A.**; Chen, C.; Rao, V.; Saibaba, A. K. (2025). *Parametric hierarchical matrix approximations to kernel matrices.* arXiv preprint arXiv:2511.03109 [math.NA]. (39 pages).
- Scott, M.; **Khan, A.**; Kilmer, M. E.; Hu, X.; Saibaba, A. K. *Hierarchical matrix approximation using tensor decompositions.* Manuscript in preparation.

Software

- **PTTK Method** — GitHub. Implementation of the PTTK method (Python).
- **Parametric Hierarchical Matrices** — GitHub. Implementation of the parametric hierarchical matrix approach (Python).
- **TensorTrain Greedy-Cross** — GitHub. Implementation of the greedy restricted cross interpolation algorithm for tensor trains (Python).

Conferences & Presentations

SIAM Conference on Computational Science and Engineering (CSE25) *March 3–7, 2025*
Fort Worth, TX
Talk: "Parametric Kernel Low-Rank Approximations Using Tensor Train Decomposition."

SIAM Conference on Mathematics of Data Science (MDS24) *October 21–25, 2024*
Atlanta, GA
Poster: "Parametric Kernel Low-Rank Approximations Using Tensor Train."

Workshop on Sparse Tensor Computations *October 18–19, 2023*
Chicago, IL
Poster: "Efficient Low-Rank Parametric Kernel Matrix Approximation Using the Tensor Train Decomposition."

Internal Talks

Randomized Numerical Analysis RTG *November 8, 2024*
Raleigh, NC
Talk: "Parametric Kernel Low-Rank Approximations Using Tensor Train."

Computing Skills

Python · C++ · C · MATLAB · Git · Linux · HPC

Teaching Experience

Teaching Assistant, MA 305: Introductory Linear Algebra and Matrices, North Carolina State University *Fall 2020*

Raleigh, NC

- Held office hours; graded assignments and exams.

Teaching Assistant, MA 242: Calculus III, North Carolina State University *2021–2022*

Raleigh, NC

Terms: Spring 2021; Fall 2021; Fall 2022.

- Led recitations; held office hours; graded assignments and exams.

Teaching Assistant, MA 121: Elements of Calculus, North Carolina State University *Spring 2022*

Raleigh, NC

- Held office hours; graded assignments and exams.